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(54) **FLUID CONNECTION ASSEMBLY**

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See application file for complete search history.

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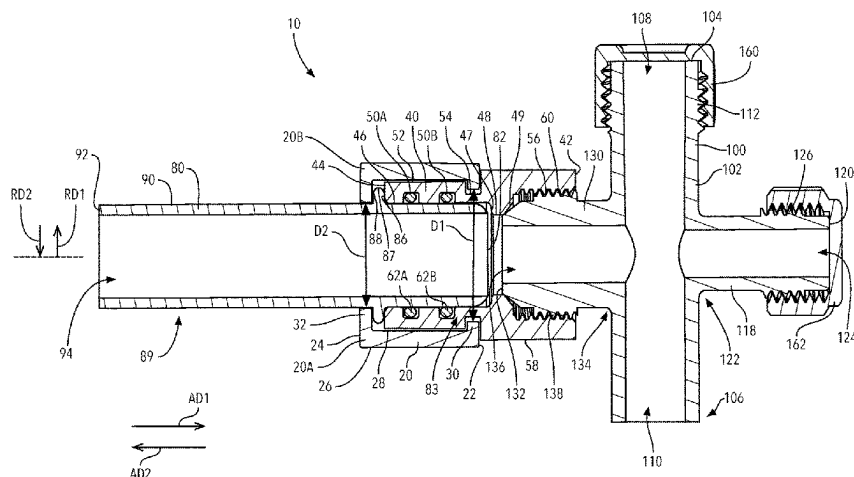
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(57) **ABSTRACT**

A fluid connection assembly, including a connector body, including a first end, a second end, a through-bore, a first radially outward facing surface including an annular groove, a first radially inward facing surface extending from the first end, and a second radially inward facing surface extending from the second end, and a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a third radially inward facing surface, a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove, and

(Continued)



a second flange extending radially inward from the third radially inward facing surface.

20 Claims, 4 Drawing Sheets

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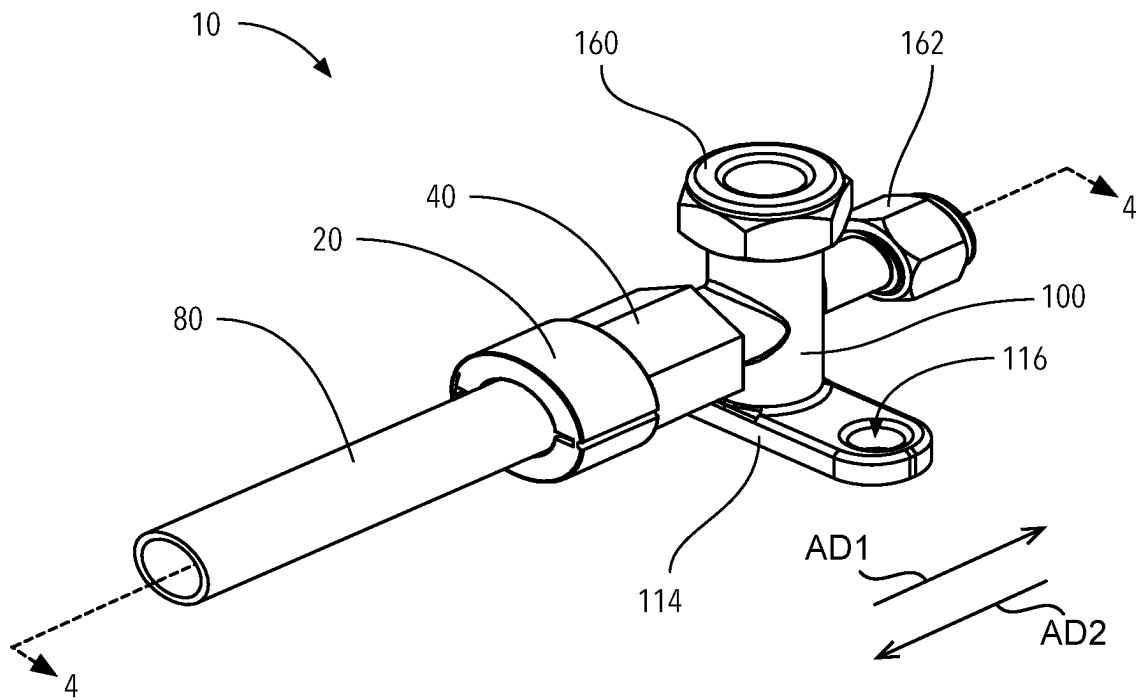


FIG. 1A

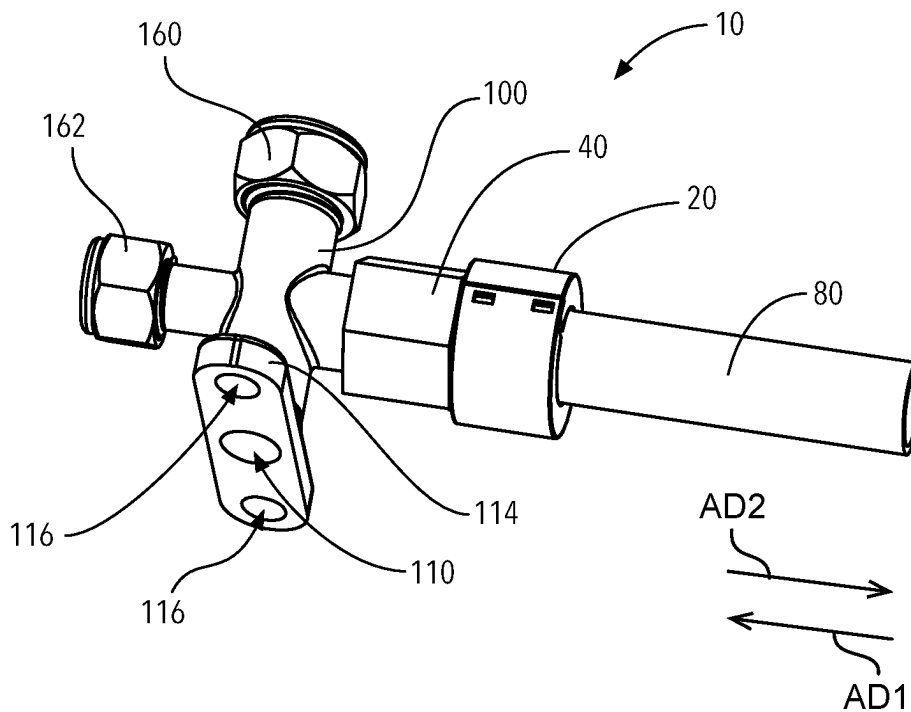


FIG. 1B

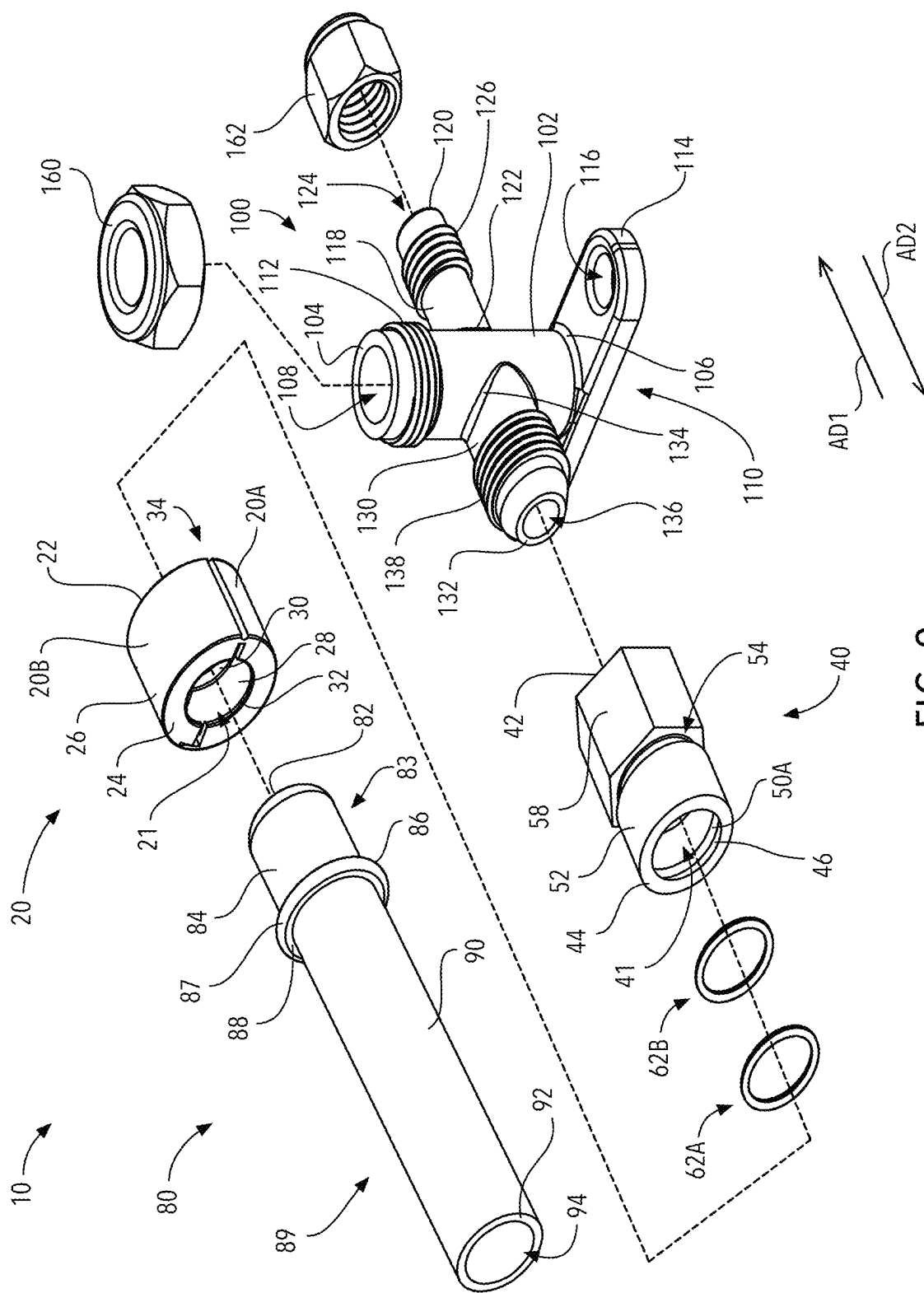


FIG. 2

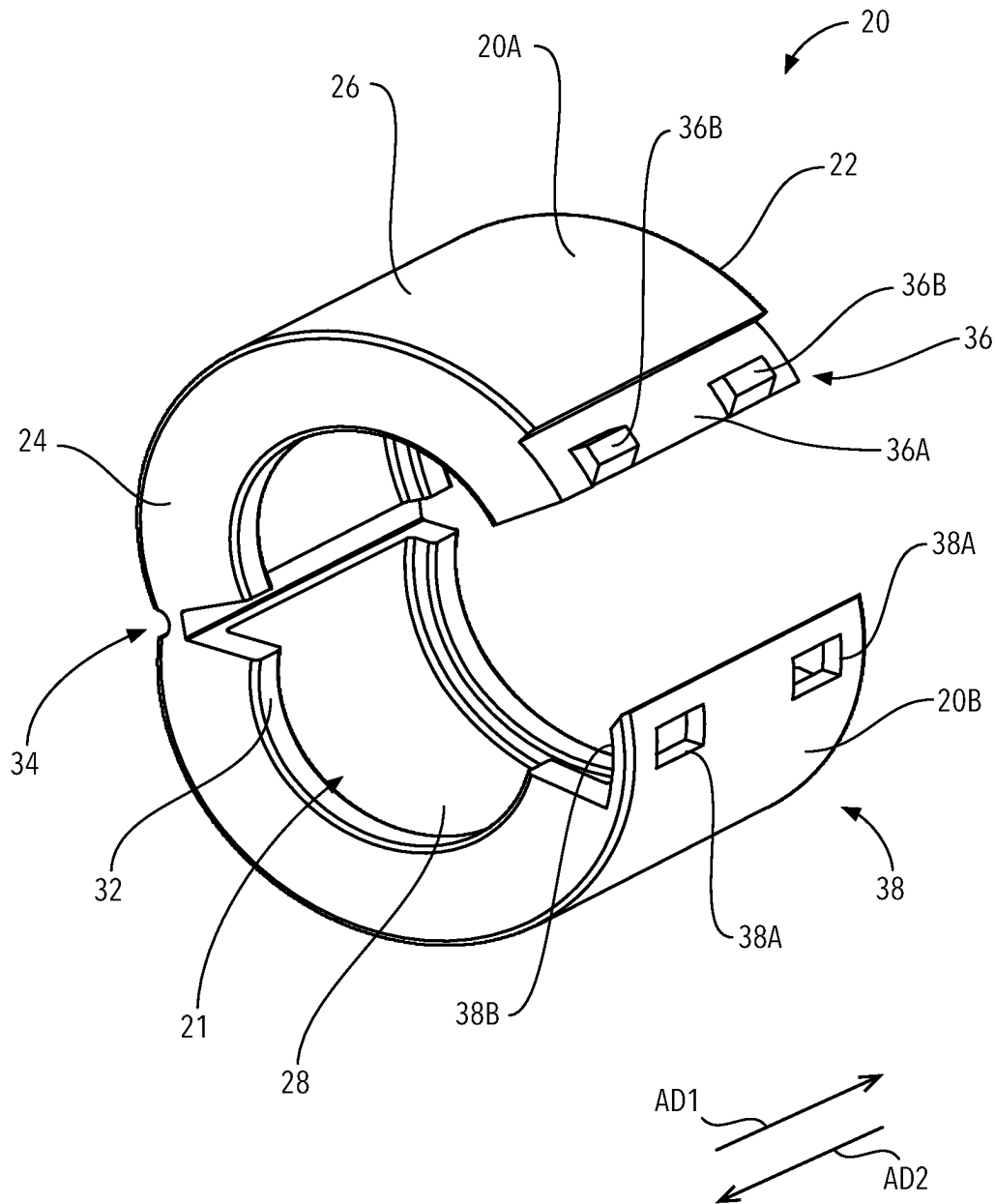


FIG. 3

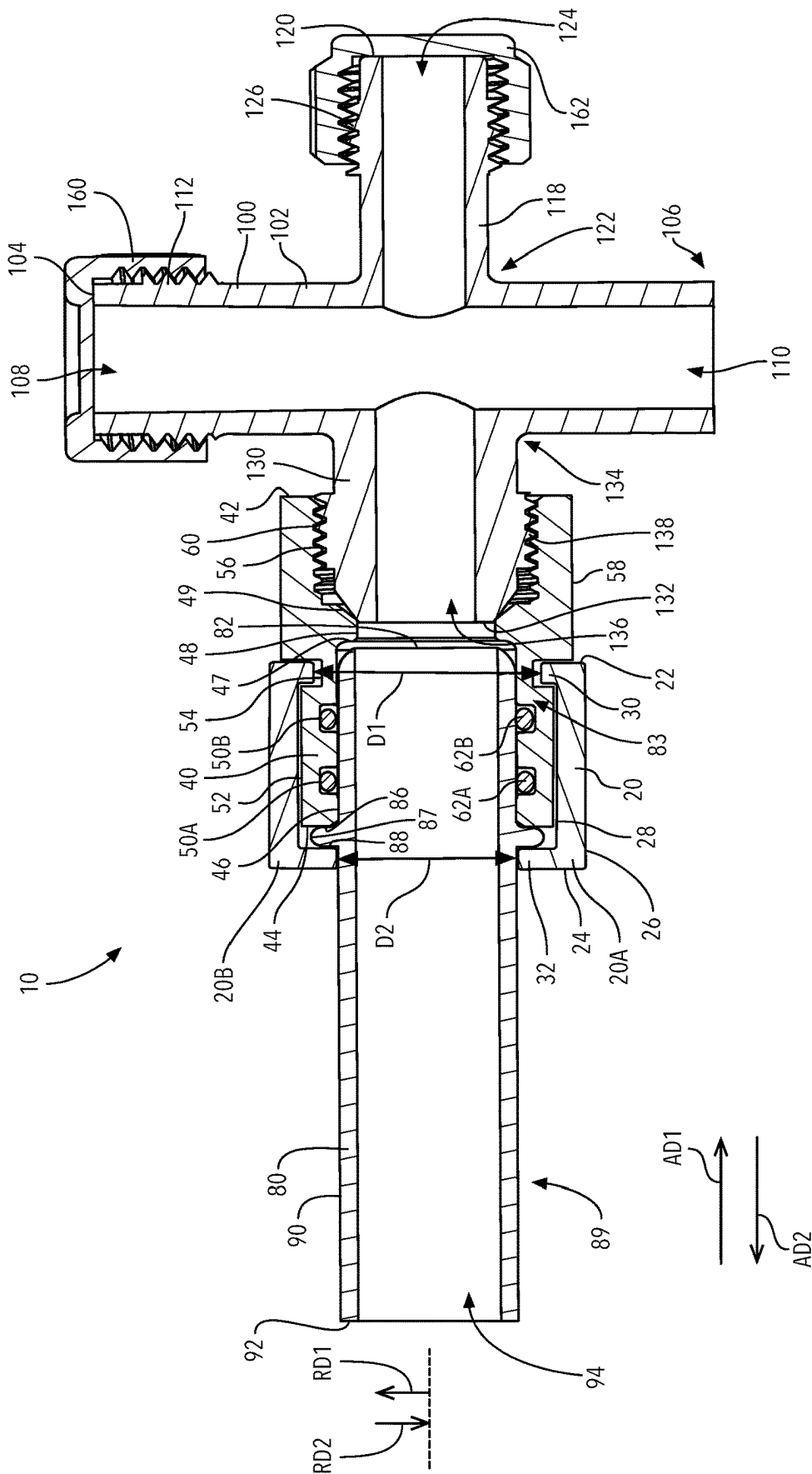


FIG. 4

1

FLUID CONNECTION ASSEMBLY**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is the U.S. National Stage Application pursuant to 35 U.S.C. § 371 of International Patent Application No. PCT/US2022/071545, filed on Apr. 5, 2022, which application claims the benefit under Articles 4 and 8 of the Stockholm Act of the Paris Convention for the Protection of Industrial Property of International Patent Application No. PCT/US2021/071953, filed on Oct. 21, 2021, and International Patent Application No. PCT/US2021/035393, filed on Jun. 2, 2021, which applications are hereby incorporated by reference herein in their entireties.

FIELD

The present disclosure relates to fluid connectors, and more particularly, to a fluid connection assembly for a refrigeration system including a connector body and a retainer that decreases the insertion force required for assembly and allows for quick assembly of components without the need for tools.

BACKGROUND

Fluid connectors, fluid connections, and fluid connection assemblies are integral components for many applications, and especially for automotive and industrial applications. Since refrigeration systems are made up of various components such as a compressor, air cooler, condenser, receiver, evaporator or chiller, and/or suction drum, fluid must be able to travel not only within each component but also between components. Fluid predominantly moves between components via flexible or rigid hoses which connect to each component by fluid connectors. Such fluid connectors typically include a retaining clip, retaining ring clip, or snap ring carried on the connector body which is adapted to snap behind a raised shoulder of a tube when the tube is fully inserted into the connector body. However, in order for the fluid connector to properly function, slots or apertures must be machined in the connector body such that the retaining clip can protrude therethrough and engage the tube, which requires extra post-process manufacturing. During the assembly process, installation of the retaining clip onto the connector body is difficult and failure to install the retaining clip properly can jeopardize the structural integrity of the retaining clip. Additionally, the force required to engage the tube into the connector body, and overcome the radial force of the retaining clip, is very large with current designs. Also, since the retaining clips are very thin and small, it is easy to lose them if dropped or misplaced. Furthermore, some connection assembly solutions take a long time to secure and require tools for the assembly process. Moreover, current connection assembly solutions do not indicate the connection state of the assembly.

Thus, there has been a long-felt need for a fluid connection assembly for a refrigeration system including a connector body and a retainer that allows for quick assembly, disassembly, eliminates the need for post-process machining, and reduces the insertion force required to assemble the fluid connector.

SUMMARY

According to aspects illustrated herein, there is provided a fluid connection assembly, comprising a connector body,

2

comprising a first end, a second end, a through-bore, a first radially outward facing surface comprising an annular groove, a first radially inward facing surface extending from the first end, and a second radially inward facing surface extending from the second end, and a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a third radially inward facing surface, a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove, and a second flange extending radially inward from the third radially inward facing surface.

In some embodiments, the first radially inward facing surface comprises at least one annular groove. In some embodiments, the at least one annular groove comprises a first annular groove, and a second annular groove spaced apart in an axial direction from the first annular groove. In some embodiments, a seal is arranged in the at least one annular groove. In some embodiments, the second radially inward facing surface comprises threading. In some embodiments, the connector body further comprises a fourth radially inward facing surface arranged between first radially inward facing surface and the second radially inward facing surface, wherein the diameter of the fourth radially inward facing surface is less than both the first radially inward facing surface and the second radially inward facing surface. In some embodiments, the connector body further comprises an axial surface connecting the first radially inward facing surface and the fourth radially inward facing surface. In some embodiments, the connector body further comprises a frusto-conical surface connecting the second radially inward facing surface and the fourth radially inward facing surface.

In some embodiments, the first flange is arranged immediately proximate the third end, and the second flange is arranged immediately proximate the fourth end. In some embodiments, the first flange comprises a first diameter, the second flange comprises a second diameter, and the first diameter is greater than the second diameter. In some embodiments, the fluid connection assembly further comprises a tube including a shoulder, wherein the retainer is operatively arranged to secure the tube to the connector body. In some embodiments, the fluid connection assembly further comprises a service valve body comprising a plurality of ports, wherein the connector body is operatively arranged to be removably connected to at least one port of the plurality of ports.

According to aspects illustrated herein, there is provided a fluid connection assembly, comprising a connector body, comprising a first end, a second end, a through-bore extending from the first end to the second end, a first radially outward facing surface comprising an annular groove, a first radially inward facing surface extending from the first end, a second radially inward facing surface extending from the second end, and a third radially inward facing surface arranged between the first radially inward facing surface and the second radially inward facing surface, the third radially inward facing surface having a diameter that is less than the diameter of both the first radially inward facing surface and the second radially inward facing surface.

In some embodiments, the fluid connection assembly further comprises a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a fourth radially inward facing surface, a first flange extending radially inward from the fourth radially inward facing sur-

3

face and operatively arranged to engage the annular groove, and a second flange extending radially inward from the fourth radially inward facing surface. In some embodiments, the first flange is arranged immediately proximate the third end, and the second flange is arranged immediately proximate the fourth end. In some embodiments, the first flange comprises a first diameter, the second flange comprises a second diameter, and the first diameter is greater than the second diameter. In some embodiments, the first radially inward facing surface comprises a first annular groove, and a second annular groove spaced apart in an axial direction from the first annular groove. In some embodiments, the second radially inward facing surface comprises threading. In some embodiments, the connector body further comprises an axial surface connecting the first radially inward facing surface and the third radially inward facing surface. In some embodiments, the connector body further comprises a frusto-conical surface connecting the second radially inward facing surface and the third radially inward facing surface.

According to aspects illustrated herein, there is provided a service port quick connect body that improves the ease of connection for industrial refrigerant applications, as well as eliminates the need for braze or solder joints. The present disclosure reduces the potential for operator induced quality failures and/or leaks, and the amount of time to connect a system.

The service port of the present disclosure comprises a quick connector body removably connectable thereto. For example, the quick connector body may be threadably engaged with the service port. The quick connector body comprises one or more O-rings and a flange to receive a plastic retainer. The tube bead abuts to the quick connect body flange and the retainer closes in on the bead and the quick connect body flange to retain the tube and prevent it from being removed. To service the system in the field, the retainer can be a replacement part provided by a service technician. Using a small pick/tool, the latch on the retainer is pried open.

These and other objects, features, and advantages of the present disclosure will become readily apparent upon a review of the following detailed description of the disclosure, in view of the drawings and appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:

FIG. 1A is a front perspective view of a fluid connection assembly;

FIG. 1B is a rear perspective view of the fluid connection assembly shown in FIG. 1A;

FIG. 2 is an exploded perspective view of the fluid connection assembly shown in FIG. 1A;

FIG. 3 is a perspective view of the retainer shown in FIG. 1A, in an unlocked state; and,

FIG. 4 is a cross-sectional view of the fluid connection assembly taken generally along line 4-4 in FIG. 1A.

DETAILED DESCRIPTION

At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.

4

Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the claims.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the example embodiments. The assembly of the present disclosure could be driven by hydraulics, electronics, pneumatics, and/or springs.

It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,” “immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.

It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a second element and a third element; or, a device comprising a second element and a third element.

Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of:” is used herein. Furthermore, as used herein, “and/or” is intended to

mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur.

It should be appreciated that the term “tube” as used herein is synonymous with hose, pipe, channel, conduit, tube end form, or any other suitable pipe flow used in hydraulics and fluid mechanics. It should further be appreciated that the term “tube” can mean a rigid or flexible conduit of any material suitable for containing and allowing the flow of a gas or a liquid.

“Fully assembled locked state” as used herein refers to the fluid connection assembly and indicates that the tube is fully connected to the service valve body and the retainer is fully connected to the service valve body to retain the tube therein. “Partially assembled locked state” as used herein refers to the fluid connection assembly and indicates that the retainer is fully connected to the service valve body but the tube is not arranged in the service valve body or the retainer. The retainer may also be described as having a “locked state” or an “unlocked state,” which indicates the state of the retainer independent of the service valve body and the tube.

Adverting now to the figures, FIG. 1A is a front perspective view of fluid connection assembly 10. FIG. 1B is a rear perspective view of fluid connection assembly 10. FIG. 2 is an exploded perspective view of fluid connection assembly 10. Fluid connection assembly 10 generally comprises retainer 20, tube 80, connector body 40, and service valve body 100. The following description should be read in view of FIGS. 1A-2.

Tube 80 comprises end 82, section 83, bead or shoulder 87, section 89, end 92, and through-bore 94. Through-bore 94 extends through tube 80 from end 82 to end 92. Section 183 is arranged between end 82 and shoulder 87 and comprises radially outward facing surface 84. Radially outward facing surface 84 includes a substantially constant diameter. In some embodiments, radially outward facing surface 84 comprises a frusto-conical taper or curvilinear surface proximate end 82 (see FIG. 4). Shoulder 87 is arranged between section 83 and section 89 and comprises surface 86 and surface 88. In some embodiments, surface 86 is an axial surface facing at least partially in axial direction AD1 and surface 88 is an axial surface facing at least partially in axial direction AD2. In some embodiments, surface 86 is a frusto-conical surface extending from the radially outward facing surface of shoulder 87 radially inward in axial direction AD1. For example, surface 86 may be a linear conical shape and increase in diameter in axial direction AD2. In some embodiments, surface 86 may comprise a linear portion and a conical or frusto-conical portion. Section 89 is arranged between shoulder 87 and end 92 and comprises radially outward facing surface 90. Radially outward facing surface 90 includes a substantially constant diameter. Tube 80 is arranged to be inserted, specifically with end 82 first, into connector body 40. Tube 80 is inserted into connector body 40 until section 83, or radially outward facing surface 84, engages seals 62A-B (see FIG. 4). Shoulder 87 engages end 44, more specifically, surface 86 abuts against and/or engages end 44, at which point retainer 20 is assembled to secure tube 80 to service valve body 100, as will be described in greater detail below. It should be appreciated that tube 80 may be any traditional tube or tube end form comprising a bead, radially outward extending protrusion or flange, or ramp profile, which extends radially outward and axially on the outer surface of the tube, to secure the tube within the connector body. In some embodiments, tube 80 comprises a metal, a polymer, and/or a ceramic.

FIG. 3 is a perspective view of retainer 20, in an unlocked state. Retainer 20 is shown in FIGS. 1A-2 in a locked state. FIG. 4 is a cross-sectional view of fluid connection assembly 10 taken generally along line 4-4 in FIG. 1A. The following description should be read in view of FIGS. 1A-4.

Retainer 20 generally comprises section 20A, section 20B, end 22, end 24, hole 21 extending from end 22 to end 24, radially outward facing surface 26, and radially inward facing surface 28. In some embodiments, section 20A is hingedly connected to section 20B, via, for example, hinge 34. In some embodiments, hinge 34 is a living hinge. In some embodiments, section 20A is removably connected to section 20B. When sections 20A and 20B are connected or retainer 20 is in the locked state (FIGS. 1A-2), hole 21 is formed therebetween; however, it should be appreciated that even in the unlocked state (FIG. 3), it could be said that each of sections 20A and 20B include a respective hole 21.

Radially inward facing surface 28 extends from end 22 to end 24 and is operatively arranged to engage radially outward facing surface 52. Radially inward facing surface 28 comprises flange 30 extending radially inward therefrom. In some embodiments, flange 30 is arranged immediately adjacent end 22. Flange 30 is operatively arranged to engage groove 54 and radially inward facing surface 28 is operatively arranged to engage radially outward facing surface 52 to connect retainer 20 to connector body 40. Flange 30 comprises diameter D1. Radially inward facing surface 28 further comprises flange 32 extending radially inward therefrom. In some embodiments, flange 32 is arranged immediately adjacent end 24. Flange 32 is operatively arranged to engage shoulder 87 to connect tube 80 to connector body 40. Flange 32 comprises diameter D2. In some embodiments, diameter D1 is greater than diameter D2.

As best seen in FIG. 4, flange 32 is spaced apart from flange 30 by a first distance. Groove 54 is spaced apart from end 44 by a second distance. Shoulder 87 comprises a width equal to a third distance. The first distance is substantially equal to the sum of the second distance and the third distance. This design results in, when retainer 20 is connected to connector body 40 and tube 80, flange 32 abutting against surface 88, as well as surface 86 abutting against end 44 (i.e., retainer 20 clamps shoulder 87 to end 44). In some embodiments, the first distance is less than the sum of the second distance and the third distance. This results in shoulder 87 and connector body 40 being squeezed or clamped together. In some embodiments, the first distance is greater than the sum of the second distance and the third distance such that shoulder 87 is engaged or in close proximity with connector body 40. Additionally, flange 30 comprises a first width and groove 54 comprises a second width, the second width being greater than the first width. As such, when retainer 20 is connected to connector body 40 without tube 80 arranged therein, retainer 20 is capable of sliding or displacing in axial direction AD1 and axial direction AD2 with respect to connector body 40. When tube 80 is properly secured to connector body 40 by retainer 20, retainer 20 is not capable of displacing in axial direction AD1 and axial direction AD2 with respect to connector body 40. This feature provides an indicator as to the connection status of fluid connection assembly 10 and will be described in greater detail below. In some embodiments, the first width is substantially equal to the second width.

As previously described, flange 32 comprises diameter D2, which is less than the diameter of shoulder 87. This design prevents shoulder 87 from being removed from connector body 40, or tube 80 from displacing in axial direction AD2 or rotating (i.e., bending moment) relative to

connector body 40. Tube 80 may still rotate in a circumferential direction with respect to connector body 40 and retainer 20.

Section 20A comprises male connector 36 and section 20B comprises female connector 38. As shown, male connector 36 on section 20A is arranged to engage female connector 38 on section 20B such that sections 20A and 20B are fixedly secured. In some embodiments, male connector 36 is hook-shaped (in a radially outward direction) and includes groove 36A and one or more radially outward extending projections 36B. Groove 36A is arranged in radially outward facing surface 26. Projections 36B extend generally radially outward in radial direction RD1 from groove 36A. In some embodiments, projections 36B comprise a tapered section near its radially outermost end operatively arranged to allow engagement of projections 36B with female connector 38, specifically, apertures 38A, to occur with greater ease.

Female connector 38 comprises one or more apertures 38A extending radially inward from radially outward facing surface 26. Apertures 38A are operatively arranged to engage projections 36B to lock section 20B to section 20A. Female connector 38 further comprises recess 38B. In some embodiments, recess 38B is arranged in radially inward facing surface 28 (i.e., extends radially outward from radially inward facing surface 28). As section 20B is displaced toward section 20A, female connector 38 engages projections 36B and displaces radially outward in radial direction RD1. Once apertures 38A are aligned with projections 36B, female connector 38 snaps back radially inward, in radial direction RD2, thereby securing section 20B to section 20A.

Service valve body 100 generally comprises section or tube 102, section or tube 118, and section or tube 130 to form a plurality of ports, for example, port 108, port 110, port 124, and port 136. Section 102, section 118, section 130 are connected such that all ports 108, 110, 124, and 136 are in fluid communication. Section 102 comprises end 104 forming port 108 and end 110 forming port 110. In some embodiments, and as shown, port 108 and port 110 are concentrically aligned. In some embodiments, port 108 is a service port and may comprise a valve stem or a stop collar (retainer) arranged therein. In some embodiments, section 102 comprises threading 112 on its radially outward facing surface proximate end 104. In some embodiments, cap 160 is threadably engaged with threading 112 to seal port 108. In some embodiments, port 110 is a middle port operatively arranged to be connected to a compressor or heat pump in a refrigeration system. In some embodiments, service valve body 100 comprises flange 114 connected to end 106. Flange 114 may comprise one or more through-holes 116 operatively arranged to secure service valve body 100 to another component (e.g., a compressor). In some embodiments, port 108 comprises a first diameter and port 110 comprises a second diameter, the second diameter being equal to the first diameter. In some embodiments, port 108 comprises a first diameter and port 110 comprises a second diameter, the second diameter being nonequal to the first diameter.

Section 118 comprises end 122 connected to section 102 and end 120 forming port 124. In some embodiments, port 124 may be a valve port and thus comprise a valve therein (e.g., a Schroeder valve). The valve in port 124 may control fluid flow within service valve body 100, for example, between ports 108, 110, and 136. In some embodiments, section 118 comprises threading 126 on its radially outward facing surface proximate end 120. In some embodiments, cap 162 is threadably engaged with threading 126 to seal port 124. In some embodiments, and as shown, port 124 is

concentrically aligned with port 136. In some embodiments, section 118 is perpendicular to section 102. In some embodiments, section 118 is non-perpendicular to section 102.

Section 130 comprises end 134 connected to section 102 and end 132 forming port 136. In some embodiments, port 136 may be a line port and is operatively arranged to be connected to a component that is filled with fluid or through which fluid flows. For example, port 136 may be connected to a refrigerant line (i.e., tube 80). In some embodiments, section 130 comprises threading 138 on its radially outward facing surface proximate end 132. In some embodiments, connector body 40 is threadably engaged with threading 138, as will be described in greater detail below. In some embodiments, section 130 is perpendicular to section 102. In some embodiments, section 130 is non-perpendicular to section 102. In some embodiments, end 132 comprises a frusto-conical or curvilinear taper, wherein the diameter of the radially outward facing surface of section 130 decreases in axial direction AD2 (see FIG. 4). In some embodiments, port 124 comprises a first diameter and port 136 comprises a second diameter, the second diameter being greater than the first diameter. In some embodiments, port 124 comprises a first diameter and port 136 comprises a second diameter, the second diameter being less than or equal to the first diameter.

Connector body 40 comprises through-bore 41 extending from end 42 to end 44, radially inward facing surface 46, groove 50, radially outward facing surface 52, groove 54, head 58, and radially inward facing surface 56. Connector body 40 is arranged to be removably and sealingly connected to service valve body 100. Connector body 40 may be screwed onto a service valve body 100, specifically, by threadably engaging threading 60 on radially inward facing surface 56 with threading 138 of section 130. Connector body 40 may be screwed onto section 130 via head 58 (e.g., using a wrench). In some embodiments, head 58 is hexagonal; however, it should be appreciated that head 58 may comprise any geometry suitable for applying torque to connector body 40. It should be appreciated that fluid connection assembly 10 may be used in various components, assemblies, and subassemblies in which fluid connection is desired, for example, refrigeration systems or compressors, or a transmission.

End 42 is connected to section 130. Connector body 40 is arranged to be connected to a component that is filled with a fluid or through which fluid flows. For example, connector body 40 may be connected to a refrigerant line (i.e., tube 80). Seals 62A-B are arranged in connector body 40. Specifically, seal 62A-B are arranged in grooves 50A-B to engage tube 80 (i.e., radially outward facing surface 84). Grooves 50A-B are arranged spaced apart axially in radially inward facing surface 46. In some embodiments, seals 62A-B are O-rings. In some embodiments, connector body 40 only comprises one groove in its radially inward facing surface 46, and one seal arranged therein. In some embodiments, radially inward facing surface 46 is a substantially cylindrical surface. End 44 is operatively arranged to engage shoulder 87, specifically, to prevent axial displacement of tube 80 is axial direction AD1.

In some embodiments, and as shown, connector body 40 comprises radially inward facing surface 48, which comprises a diameter that is less than the diameter of radially inward facing surface 46. Radially inward facing surface 48 is connected radially inward facing surface 146 via surface 47. In some embodiments, surface 47 is a substantially axial facing surface and faces axial direction AD2. In some embodiments, surface 47 acts as a stop or seat for tube 80

thereby preventing tube **80** from being displaced in axial direction AD1 with respect to connector body **40**. In some embodiments, and as shown, connector body **40** further comprises radially inward facing surface **49** which connects radially inward facing surface **48** with radially inward facing surface **56**. Surface **49** acts as a stop or seat for section **130**. In some embodiments, radially inward facing surface **49** is a frusto-conical surface that increases in diameter in axial direction AD1 (see FIG. 4). In some embodiments, radially inward facing surface **56** comprises a diameter that is greater than radially inward facing surface **46**. In some embodiments, radially inward facing surface **56** comprises a diameter that is less than or equal to radially inward facing surface **46**.

Groove **54** is arranged in radially outward facing surface **52**. Groove **54** is arranged axially between end **44** and end **42**. Groove **54** comprises a radially outward facing surface and two axial facing surfaces. The radially outward facing surface of groove **54** comprises a diameter that is less than the diameter of radially outward facing surface **52**. In some embodiments, groove **54** is arranged axially between and spaced apart from end **44** and head **58**. In some embodiments, and as shown, groove **54** is arranged immediately adjacent head **58**. Groove **54** is operatively arranged to engage with flange **30** to connect retainer **20** to connector body **40**.

Groove **54** comprises a width that is greater than the width of flange **30**. In the partially assembled locked state, when retainer **20** is connected to connector body **40**, without tube **80** properly connected or present at all, flange **30** and thus retainer **20** is displaceable in axial direction AD1 and axial direction AD2 with respect to groove **54** and thus connector body **40** (i.e., flange **30** can slide back and forth within groove **54**). This provides an indication that fluid connection assembly **10** is not properly assembled. When retainer **20** is connected to connector body **40**, with tube **80** properly connected, flange **30** and thus retainer **20** is not displaceable (or is not displaceable as much) in axial direction AD1 and axial direction AD2 with respect to groove **54** and thus connector body **40** (i.e., flange **30** abuts against the left surface of groove **54** as shown in FIG. 4). In some embodiments, connector body **40** comprises a metal, a polymer, and/or a ceramic.

To assemble fluid connection assembly **10**, connector body **40** is connected to service valve body **100**, specifically, section **130**. For example, connector body **40** may be screwed onto section **130** such that threading **60** is engaged with threading **138**. In some embodiments, end **132** (or its frusto-conical taper) is engaged with radially inward facing surface **49**. In some embodiments, end **132** is spaced apart from radially inward facing surface **49**. Tube **80** is inserted in axial direction AD1, with end **82** first, into connector body **40**. Radially outward facing surface **84** engages seals **62A-B** and section **83** is arranged inside of connector body **40** proximate radially inward facing surface **46**. Shoulder **87** engages end **44** of connector body **40**, specifically, surface **86** abuts against end **44**. In some embodiments, and as shown, shoulder **87** resides completely outside of connector body **40**. In some embodiments, end **82** abuts against surface **47**. In some embodiments, end **82** is spaced apart from surface **47**. Then, retainer **20** is secured over both connector body **40** and tube **80**. Specifically, sections **20A** and **20B** are arranged over connector body **40** such that flange **30** is at least partially aligned with groove **54** and flange **32** is at least partially aligned with surface **88**. Sections **20A** and **20B** are displaced radially inward toward each other (i.e., in radial direction RD2) until female connector **38** fully engages male

connector **36** and retainer **20** is in the locked state. Additionally, flange **32** engages tube **80**, specifically radially outward facing surface **90**, thereby preventing the ingress of foreign materials into fluid connection assembly **10**.

In the fully assembled locked state, flange **30** engages groove **54**, radially inward facing surface **28** engages radially outward facing surface **52** and flange **32** engages shoulder **87** of tube **80**. In some embodiments, in the fully assembled locked state, flange **30** abuts against the left surface of groove **54**, flange **32** abuts against surface **88**, and surface **86** abuts against end **44**. In some embodiments, in the fully assembled locked state, flange **30** abuts against the radially outward facing surface of groove **54**, thus preventing radial or circumferential displacement of retainer **20** with respect to connector body **40**. In some embodiments, flange **32** engages radially outward facing surface **90**. The engagement of tube **80** with connector body **40** prevents axial displacement of tube **80** in axial direction AD1 and the engagement of retainer **20** with connector body **40** and tube **80** prevents axial displacement of tube **80** in axial direction AD2, as well as radial directions RD1 and RD2, relative to connector body **40**.

To disassemble, female connector **38** is displaced radially outward in radial direction RD1 with respect to male connector **36** until apertures **38A** disengages projections **36B**. Sections **20A** and **20B** are then separated to disengage flange **32** from shoulder **87**, at which point tube **80** can be removed from connector body **40**.

It will be appreciated that various aspects of the disclosure above and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

LIST OF REFERENCE NUMERALS

10	Fluid connection assembly
20	Retainer
20A	Section
20B	Section
21	Hole
22	End
24	End
26	Radially outward facing surface
28	Radially inward facing surface
30	Flange
32	Flange
34	Hinge
36	Male connector
36A	Groove
36B	Projection(s)
38	Female connector
38A	Aperture(s)
38B	Recess
40	Connector body
41	Through-bore
42	End
44	End
46	Radially inward facing surface
47	Axial surface or seat
48	Radially inward facing surface
49	Radially inward facing surface
50A	Groove
50B	Groove

11

52 Radially outward facing surface
 54 Groove
 56 Radially inward facing surface
 58 Head
 60 Threading
 62A Seal
 62B Seal
 80 Tube
 82 End
 83 Section
 84 Radially outward facing surface
 86 Surface
 87 Shoulder or bead
 88 Surface
 89 Section
 90 Radially outward facing surface
 92 End
 94 Through-bore
 100 Service valve body
 102 Section or tube
 104 End
 106 End
 108 Port or service port
 110 Port or middle port
 112 Threading
 114 Flange
 116 Hole(s)
 118 Section or tube
 120 End
 122 End
 124 Port or valve port
 126 Threading
 130 Section or tube
 132 End
 134 End
 136 Port or line port
 138 Threading
 160 Cap
 162 Cap
 AD1 Axial direction
 AD2 Axial direction
 RD1 Radial direction
 RD2 Radial direction
 D1 Diameter
 D2 Diameter
 What is claimed is:
 1. A fluid connection assembly, comprising:
 a connector body, comprising:
 a first end;
 a second end;
 a through-bore;
 a first radially outward facing surface comprising an
 annular groove;
 a first radially inward facing surface extending from the
 first end; and
 a second radially inward facing surface extending from
 the second end; and
 a retainer operatively arranged to be removably connected
 to the connector body, the retainer including:
 a third end;
 a fourth end;
 a first section comprising a male connector component;
 a second section displaceable with respect to the first
 section, the second section comprising a female
 connector component operatively arranged to engage
 the male connector component to fixedly secure the
 second section to the first section;

12

a third radially inward facing surface;
 a first flange extending radially inward from the third
 radially inward facing surface and operatively
 arranged to engage the annular groove, the first
 flange comprising a first innermost diameter; and
 a second flange extending radially inward from the
 third radially inward facing surface, the second
 flange comprising a second innermost diameter, less
 than the first innermost diameter.
 2. The fluid connection assembly as recited in claim 1,
 wherein the first radially inward facing surface comprises at
 least one annular groove.
 3. The fluid connection assembly as recited in claim 2,
 wherein the at least one annular groove comprises:
 a first annular groove; and
 a second annular groove spaced apart in an axial direction
 from the first annular groove.
 4. The fluid connection assembly as recited in claim 2,
 wherein a seal is arranged in the at least one annular groove.
 5. The fluid connection assembly as recited in claim 1,
 wherein the second radially inward facing surface comprises
 threading.
 6. The fluid connection assembly as recited in claim 1,
 wherein the connector body further comprises a fourth
 radially inward facing surface arranged between the first
 radially inward facing surface and the second radially
 inward facing surface, wherein a diameter of the fourth
 radially inward facing surface is less than both the first
 radially inward facing surface and the second radially
 inward facing surface.
 7. The fluid connection assembly as recited in claim 6,
 wherein the connector body further comprises an axial
 surface connecting the first radially inward facing surface
 and the fourth radially inward facing surface.
 8. The fluid connection assembly as recited in claim 1,
 wherein:
 the first flange is arranged immediately proximate the
 third end; and
 the second flange is arranged immediately proximate the
 fourth end.
 9. The fluid connection assembly as recited in claim 1,
 further comprising a tube including a shoulder, wherein the
 retainer is operatively arranged to secure the tube to the
 connector body.
 10. The fluid connection assembly as recited in claim 9,
 wherein in a connected state, the shoulder and the connector
 body are squeezed together by the retainer.
 11. The fluid connection assembly as recited in claim 1,
 further comprising a service valve body comprising a plu-
 rality of ports, wherein the connector body is operatively
 arranged to be removably connected to at least one port of
 the plurality of ports.
 12. A fluid connection assembly, comprising:
 a connector body, comprising:
 a first end;
 a second end;
 a through-bore;
 a first radially outward facing surface comprising an
 annular groove;
 a first radially inward facing surface extending from the
 first end; and
 a second radially inward facing surface extending from
 the second end; and
 a retainer operatively arranged to be removably connected
 to the connector body, the retainer including:
 a third end;
 a fourth end;

13

a second radially outward facing surface;
 a third radially inward facing surface;
 a first section comprising a male connector component
 fixedly secured to the second radially outward facing
 surface;
 a second section displaceable with respect to the first
 section, the second section comprising a female
 connector component including a through-hole
 operatively arranged to engage the male connector
 component to fixedly secure the second section to the
 first section;
 a first flange extending radially inward from the third
 radially inward facing surface and operatively
 arranged to engage the annular groove; and
 a second flange extending radially inward from the
 third radially inward facing surface.

13. The fluid connection assembly as recited in claim 12,
 wherein:

the male connector component extends radially outward
 from the second radially outward facing surface; and
 the second section is hingedly connected to the first
 section.

14. The fluid connection assembly as recited in claim 12,
 wherein:

the first flange is arranged immediately proximate the
 third end; and
 the second flange is arranged immediately proximate the
 fourth end.

15. The fluid connection assembly as recited in claim 12,
 wherein:

the first flange comprises a first diameter;
 the second flange comprises a second diameter; and
 the first diameter is greater than the second diameter.

16. The fluid connection assembly as recited in claim 12,
 wherein the first radially inward facing surface comprises:

a first annular groove; and
 a second annular groove spaced apart in an axial direction
 from the first annular groove.

17. The fluid connection assembly as recited in claim 12,
 wherein the second radially inward facing surface comprises
 threading.

18. The fluid connection assembly as recited in claim 12,
 wherein the connector body further comprises:

14

a third radially inward facing surface arranged between
 the first radially inward facing surface and the second
 radially inward facing surface, the third radially inward
 facing surface having a diameter that is less than a
 diameter of both the first radially inward facing surface
 and the second radially inward facing surface; and
 an axial surface connecting the first radially inward facing
 surface and the third radially inward facing surface.

19. The fluid connection assembly as recited in claim 18,
 wherein the connector body comprises a head arranged at
 the second end.

20. A fluid connection assembly, comprising:

a connector body, comprising:

a first end;
 a second end;
 a through-bore;
 a first radially outward facing surface comprising an
 annular groove;
 a first radially inward facing surface extending from the
 first end; and
 a second radially inward facing surface extending from
 the second end;

a retainer operatively arranged to be removably connected
 to the connector body, the retainer including:

a third end;
 a fourth end;
 a first section comprising a male connector component;
 a second section displaceable with respect to the first
 section, the second section comprising a female
 connector component operatively arranged to engage
 the male connector component to fixedly secure the
 second section to the first section;
 a third radially inward facing surface;
 a first flange extending radially inward from the third
 radially inward facing surface and operatively
 arranged to engage the annular groove; and
 a second flange extending radially inward from the
 third radially inward facing surface; and

a tube including a shoulder, wherein the retainer is opera-
 tively arranged to secure the tube to the connector
 body;

wherein in a connected state, the shoulder and the con-
 nector body are squeezed together by the retainer.

* * * * *